

Execution of Modal Triplet Theory II: Flavor, CKM, Neutral, and Higgs Status after Profile-Standard SM Closure

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Version 3

Abstract

We report the current MTT flavor, CKM, neutral, and Higgs execution. The 27-by-27 qutrit–Weyl architecture, finite-replay charged Yukawa rows, selected CKM rows, shared electroweak primitive, direct Higgs/threshold row, and completed 96-dimensional finite Dirac operator support embedded renormalized-Standard-Model equivalence at the adopted profile standard. CKM closes at the prediction-with-uncertainty criterion, with maximum displacement 2.3565×10^{-4} standard deviations. In the neutral sector, the internal 9/9 dimensionless response and its one-scale no-go remain valid. A new two-primitive Dirac profile uses the measured oscillation splittings to emit the complete normal-ordering mass spectrum, Dirac Yukawa matrix, and flavor mass-squared matrix. This is profile calibration, not MTT selection of the splittings, ordering, Dirac ontology, or covariance. The selected finite one-form space contains three raw scalar doublets, while the q79/proto-spinor projector isolates one exact Higgs doublet. Strict numerical source promotion and absolute neutral ontology remain open.

Version 3 Revision Note

Supersedes *Execution of Modal Triplet Theory II: Flavor, CKM/PMNS, and Higgs Sector on the CY Corner*, version 2.

Reason Fitted Calabi–Yau flavor matrices, seesaw assumptions, and a few-TeV Higgs benchmark were reported too strongly and no longer represented the current finite-operator calculation. The earlier revision also stopped before the two-primitive neutral profile, completed finite Dirac operator, and one-Higgs projection.

Resolution Version 3 records the locked charged-flavor, CKM, electroweak, Higgs/threshold, neutral-profile, and finite-triple packets while preserving the one-scale/nil-subtraction obstruction and all strict-source guards.

Retained result Flavor and Higgs data close embedded renormalized-SM equivalence at the declared one-shared-primitive/profile standard.

Remaining boundary A selected source for the empirical profile, neutral ordering and ontology, uncertainty transport, and absolute normalization remains beyond that standard.

1 Closure standard

The accepted result is embedded renormalized-SM equivalence on the selected branch at the one-shared-physical-primitive/profile standard. It imports the standard SM quantization and uses a declared measured-input profile. It does not derive all empirical values from MTT, prove unique observed-branch selection, or establish a zero-primitive/no-knob theory.

The older Execution II matrices were constructed with adjustable local texture entries, phases, scale choices, and benchmark masses. The first correction pass made their real-matrix diagonalizations arithmetically reproducible and removed an unsupported CP claim. The present revision goes further: those matrices are classified as historical profile fits and are no longer the authority for flavor closure.

2 Charged flavor and the 27-by-27 operator

The current charged-sector authority contains:

- the locked 27-by-27 qutrit–Weyl/minimal matrix ledger;
- the counted AH-equivalent HYM/projective lane, closed 8/8;
- the selected charged Yukawa basis and finite-replay magnitude rows;
- the explicit profile-valued 96×96 finite Dirac operator; and
- the selected common-scheme threshold and mass transport with covariance.

These objects establish that the selected branch can reproduce the charged renormalized-SM data coherently through one operator architecture.

The nine charged Yukawa rows close at the accepted finite projected replay/profile standard. The corresponding finite Dirac operator satisfies self-adjointness, oddness, KO-dimension-six reality, order zero, and order one on the completed finite module. These statements do not make the measured charged masses zero-primitive predictions: the common measured source profile enters their numerical values. The matrix, basis maps, and finite-triple identities are source-side structural achievements; the magnitudes remain profile-calibrated outputs until a selected continuum action emits the source coordinates.

3 CKM and weak CP violation

Three selected Π_{CKM} source rows are retained. They use neither observed CKM values nor target residual fitting as selectors. Against the frozen declared CKM uncertainty profile, their maximum displacement is

$$2.3564680386 \times 10^{-4} \text{ standard deviations.} \quad (3.1)$$

The strict-upgrade row U4 therefore closes at the prediction-with-uncertainty-profile standard. Exact equality to a moving global fit central estimator is neither obtained nor required, and no empirical residual correction is introduced.

This CKM result supersedes the legacy printed real Yukawa matrices. Those real matrices could reproduce mixing magnitudes but could not establish a nonzero Jarlskog invariant. Weak-sector CP now belongs to the selected complex branch and Π_{CKM} execution. It must not be confused with strong-CP closure, which remains open.

4 Neutral and PMNS sector

4.1 Internal response and the one-scale no-go

The selected same-source neutral construction has advanced beyond the old benchmark seesaw. Its complete internal dimensionless response is

$$a_{\text{int}} = 0.34195899479289005, \quad dY_\nu = I_3 + X_3, \quad (4.1)$$

with

$$H_{1,\nu} = \begin{pmatrix} -2a & 0 & -2a \\ 0 & -2a & -2a \\ -2a & -2a & 0 \end{pmatrix}. \quad (4.2)$$

All nine $H_{1,\nu}$ rows and all seven provenance fields are selected: six entries equal $-2a$ and three are exact zeros. The associated selected two-representative orbit has diagonal coefficient 1, cyclic-shift coefficients $3/2 \pm i\sqrt{3}/2$, and Hermitian spectrum $[1, 4, 7]$.

This is not yet a dimensionful neutrino Yukawa or mass matrix. A common scale cannot alter eigenvalue ratios. Nil subtraction gives $[0, 3, 6]$ and therefore the invariant ratio

$$r_{\text{direct}} = \frac{3}{6} = \frac{1}{2},$$

whereas the stored normal-ordering oscillation postcheck is $r_{\text{post}} = 0.02980501393$. The scale-only completion is therefore falsified. This excludes identifying the raw $[1, 4, 7]$ orbit with the physical mass-squared spectrum by a common scale; it does not exclude a declared measured-profile calibration.

4.2 Two-primitive Dirac profile execution

At the declared normal-ordering Dirac profile, impose the nil-boundary condition $m_{\text{lightest}} = 0$ and use the two measured coordinates

$$\Delta m_{21}^2 = 7.49 \times 10^{-5} \text{ eV}^2, \quad \Delta m_{31}^2 = 2.513 \times 10^{-3} \text{ eV}^2.$$

Writing $r = \Delta m_{21}^2 / \Delta m_{31}^2$ and $c_k = \cos(\phi_\nu + 2\pi k/3)$, with c_{max} and c_{min} the extreme orbit values, the analytic profile inversion is

$$\phi_\nu = \arctan\left(\frac{\sqrt{3}r}{2-r}\right), \quad A_\nu = \frac{\Delta m_{31}^2}{c_{\text{max}} - c_{\text{min}}}.$$

It gives

$$\phi_\nu = 0.02619638630, \quad A_\nu = 0.001650933111 \text{ eV}^2$$

and the complete mass profile

$$(m_1, m_2, m_3) = (0, 0.00865447861, 0.05012983144) \text{ eV}, \quad \sum_i m_i = 0.05878431005 \text{ eV}.$$

With the locked PMNS replay, charged-lepton basis, and $U_R = I$ right-handed mass convention, the execution fills all Dirac mass, Yukawa, and flavor mass-squared matrix rows. The finite 96×96 profile Dirac operator uses this same neutral matrix.

This is a complete two-primitive measured-profile value closure. The two oscillation splittings are calibration inputs, not predictions. MTT has not selected normal ordering, Dirac rather than Majorana ontology, the right-handed basis convention, or the missing covariance. The result therefore closes a reproducible profile lane without closing strict U5.

4.3 Conditional one-primitive reduction

For the independently selected retarded Lens pair $16 \rightarrow 15$, Dedekind reciprocity gives the exact mixed residue

$$R_{15,16} = \frac{1}{240}.$$

If a physical Bismut–Freed/APS determinant-line theorem proves that this mixed term survives the shared-circle cancellation with the required spin, orientation, and counterterm conventions, then

$$\phi_\nu = 2\pi R_{15,16} = \frac{\pi}{120}.$$

That conditional result would determine the oscillation ratio and reduce the two measured neutral coordinates to one scale coordinate. The arithmetic identity is exact; the physical determinant-family identification is open, so the near agreement with the two-primitive profile is a postcheck rather than a prediction.

A separate 11-dimensional attenuation candidate can be compressed algebraically to one heat-semigroup factor. Native MTT has the $4 + (1 + 2 + 3) = 10$ census, however, and no theorem places the physical neutral operator on the extra circle lift. The native ten-dimensional version misses the profile amplitude by a factor 448^2 . This is a useful source discriminator, not a promoted mass law.

The old fitted PMNS, seesaw, and Majorana benchmarks remain withdrawn as predictions. The current profile is retained only with its explicit two measured inputs and ontology choices.

5 Higgs status

At the adopted profile standard, P_{EW} is counted once as the one shared physical primitive, with zero Higgs-specific free parameters. The direct $K_{\text{threshold}}.\Omega_H.\lambda$ row and its row-purpose/formula bridge are locked, and the common SMDR transport carries the Higgs row in the selected precision object.

The completed finite triple sharpens the structural Higgs statement. Exhaustive execution of all 676 finite-algebra one-form basis pairs gives a raw 12-real-dimensional scalar space, organized as three rank-four doublet modules. Thus the finite triple alone does not automatically have the one-doublet Standard Model Higgs sector. The selected q79/proto-spinor alignment rule is an exact rank-four submodule projector, with numerical residual 6.15×10^{-15} , and removes the other eight real scalar directions. This closes the one-Higgs module at the finite structural tier.

Together, these results close the Higgs coordinate for embedded renormalized-SM equivalence and the one-doublet carrier structure. They do not make λ_H an internally emitted no-knob value: its physical value still uses the measured profile and shared normalization primitive. Absolute spectral moments and field normalization remain open. The former $\tan \beta = 10, 5$ TeV supersymmetric matching benchmark and its one-loop Higgs pole-mass corridor are withdrawn.

6 Sector status table

Sector	Closed result	Remaining stronger obligation
Charged flavor	27-by-27 structure, finite-replay magnitudes, profile D_F	continuum source emission of measured profile
CKM	selected prediction profile; U4 closed	no stronger central-estimator identity required
Neutral/PMNS	internal 9/9 response and two-primitive Dirac profile	determinant-line source, ordering/ontology, covariance
Higgs	one-Higgs projector, direct K row, profile coordinate	strict normalization and spectral moments

7 Interpretation

The charged and neutral executions now have a parallel logical form. In each case the finite carrier and matrix maps are explicit, and a declared measured profile produces all physical rows. The charged lane is part of the 12/12 embedded-SM baseline; the neutral lane is a two-coordinate extension. Neither profile becomes a source prediction merely because its downstream matrix is fully determined.

There is nevertheless genuine reduction of arbitrariness in the structural layer. The same finite module carries charged and neutral Dirac entries, CKM is constrained by selected complex source rows, and the scalar one-form space is reduced to one exact Higgs doublet by a pre-existing q_{79} /proto-spinor projector. The next decisive advance must occur before profile insertion: a selected continuum action must emit the flavor magnitudes, neutral phase and scale, and Higgs normalization with a common provenance certificate.

8 Conclusion

Execution II now records the actual frontier. Charged flavor, CKM, and Higgs participate in the closed 12/12 embedded renormalized-SM equivalence theorem at the adopted one-shared-primitive/profile standard. The CKM strict upgrade is also closed at the correct uncertainty-profile criterion.

The result is not a derivation of all fermion masses from zero empirical inputs. Charged Yukawa magnitudes close at finite-replay/profile tier, and the neutral extension closes numerically with two measured oscillation coordinates under explicit Dirac and ordering choices. Physical neutrino completion is impossible by a common scaling of the raw [1, 4, 7] response, while the conditional Dedekind reduction still lacks its determinant-line source theorem. This separation preserves the genuine matrix, finite-triple, one-Higgs, and CKM achievements without promoting calibrated values into zero-primitive predictions.

Computational Evidence and Reproducibility

This execution paper directly reports the flavor, CKM, neutral, Higgs, finite-matrix, and common-scheme packets listed below. The anomaly row is a neighboring consistency check rather than a source for the flavor values. Profile replay and certified numerics retain their declared tiers, and the strict no-knob upgrade remains open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Corpus-state cross-checks.

- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] P. Nero, *Modal Triplet Theory: Admissibility, Encodings, and the Structure of Physical Description*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255621>
- [2] P. Nero, *Modal Triplet Theory: Foundation*, Zenodo preprint, September 2025. <https://doi.org/10.5281/zenodo.16949762>
- [3] P. Nero, *Fixed Points I–VI: Complete Coherence Spine*, Zenodo preprints, August 2025. <https://doi.org/10.5281/zenodo.16948748>

- [4] P. Nero, *The Projection–Admissibility Principle: Structural Constraints on Effective Physical Description*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255838>
- [5] P. Nero, *Closure and Inevitability in Modal Triplet Theory*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255510>
- [6] P. Nero, *Coherence Capacity as the Fundamental Resource of Effective Physics*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18255905>
- [7] P. Nero, *Dynamics of Coherence Capacity: Transport, Concentration, and Exhaustion*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18256048>
- [8] P. Nero, *Modal Triplet Theory: From MTT to Quantum Mechanics*, Zenodo preprint, September 2025. <https://doi.org/10.5281/zenodo.17074246>
- [9] P. Nero, *From MTT to Quantum Field Theory*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17068816>
- [10] P. Nero, *Modal Triplet Theory: From MTT to General Relativity*, Zenodo preprint, October 2025. <https://doi.org/10.5281/zenodo.16950597>
- [11] P. Nero, *Modal Triplet Theory: From MTT to a UV-Finite, Unitary Quantum Gravity*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17077671>
- [12] P. Nero, *Measurement as Disturbance and Stabilization in Modal Triplet Theory*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17177404>
- [13] P. Nero, *Projection, Probability, and Irreversibility: Shadow Bridges Between Measurement, Black Holes, and Cosmology in Modal Triplet Theory*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18256408>
- [14] P. Nero, *Modal Fixed Points, Bell’s Beables, and the Limits of Factorization*, Zenodo preprint, 2025. <https://doi.org/10.5281/zenodo.17076300>
- [15] P. Nero, *Temporal Bell Inequalities and Global Consistency in Modal Triplet Theory*, Zenodo preprint, August 2025. <https://doi.org/10.5281/zenodo.18208884>
- [16] P. Nero, *From Modal Triplet Theory to Indivisible Stochastic Processes: A First-Principles, Fully Rigorous Derivation*, Zenodo preprint, January 2026. <https://doi.org/10.5281/zenodo.18254862>